

ggplot2 internals · aesthetics · geoms

Day 7 - Wednesday, May 27, 2026

i Today: Wednesday May 27

Before class

- Perusall Assignment: [DC §6.1–6.5](#)

Plan for today

- The grammar of graphics: a vocabulary for *any* plot
- Aesthetic mappings vs. fixed properties (and a common bug)
- Geoms in depth: layering, local vs. global, multiple data
- Statistical transformations: where do bar heights come from?
- Position adjustments: stacking, dodging, jittering
- In-class activity

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 9](#) · [DC §6](#) · [ggplot2 reference](#)

Quick recap

We've been making ggplots for a while now. You know how to:

- Pass `data` and `aes()` to `ggplot()`
- Add a geom like `geom_point()`, `geom_bar()`, `geom_histogram()`
- Map variables to `x`, `y`, `color`, `fill`

Today we're going to **slow down** and look at what's actually happening under the hood.

💡 Tip

The goal of today is to give you the tools to be able to build any plot you can describe.

The grammar of graphics

A vocabulary for plots

The DC chapter introduces five words. These will come up a lot.

Table 1: Five words to describe any data graphic.

Term	What it is
frame	The relationship between <i>position</i> and <i>data</i> . The coordinate system + the variables on each axis.
glyph	The basic graphical unit. One glyph = one case in your data frame.
aesthetic	Any graphical attribute of a glyph: position, size, color, shape, transparency, ...
scale	The rule that translates a <i>variable value</i> into an <i>aesthetic value</i> .
guide	What helps the viewer translate back: axes, tick marks, legends, color bars.

One sentence, five words

💡 Tip

A graphic is a collection of **glyphs** sitting inside a **frame**, where each glyph's **aesthetics** are controlled by **scales** that map variables to graphical attributes, and **guides** help the viewer read the scales backward.

Today we focus on the first three; tomorrow we'll deal more with scales and guides.

Why this matters

Once you have the vocabulary, debugging gets easier:

- “My color isn’t working” → is it a *mapping* (inside `aes()`) or a *fixed property* (outside)?
- “I want different lines per group” → I need a discrete variable mapped to an aesthetic that *splits* the geom (color, linetype, group).

- “My bar chart shows counts but I want proportions” → I need to override the **stat**.

Every confusion you’ve had with ggplot maps onto one of these concepts.

A dataset for today

The storms dataset

Built into the tidyverse (via `dplyr`). Records of every named Atlantic storm since 1975, drawn from NOAA’s HURDAT2 reanalysis.

`storms`

```
# A tibble: 20,778 x 13
  name   year month   day hour   lat   long status   category wind pressure
  <chr> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <fct>      <dbl> <int>    <int>
1 Amy   1975     6    27     0  27.5 -79 tropical d~    NA     25     1013
2 Amy   1975     6    27     6  28.5 -79 tropical d~    NA     25     1013
3 Amy   1975     6    27    12  29.5 -79 tropical d~    NA     25     1013
4 Amy   1975     6    27    18  30.5 -79 tropical d~    NA     25     1013
5 Amy   1975     6    28     0  31.5 -78.8 tropical d~    NA     25     1012
6 Amy   1975     6    28     6  32.4 -78.7 tropical d~    NA     25     1012
7 Amy   1975     6    28    12  33.3 -78 tropical d~    NA     25     1011
8 Amy   1975     6    28    18   34  -77 tropical d~    NA     30     1006
9 Amy   1975     6    29     0  34.4 -75.8 tropical s~    NA     35     1004
10 Amy  1975     6    29     6   34  -74.8 tropical s~    NA     40     1002
# i 20,768 more rows
# i 2 more variables: tropicalstorm_force_diameter <int>,
#   hurricane_force_diameter <int>
```

Variables we’ll use

Table 2: Selected variables in `storms`.

Variable	Description
name, year, month, day, hour	When the observation was taken
lat, long	Where the storm was
status	tropical depression, tropical storm, hurricane, ...
category	Saffir-Simpson scale (1–5), NA for non-hurricanes
wind	Maximum sustained wind speed, knots

Variable	Description
pressure	Central pressure, millibars (lower = stronger)

Each row is a **6-hour observation** of one storm, so single hurricane shows up many times as it moves and intensifies.

Aesthetic mappings

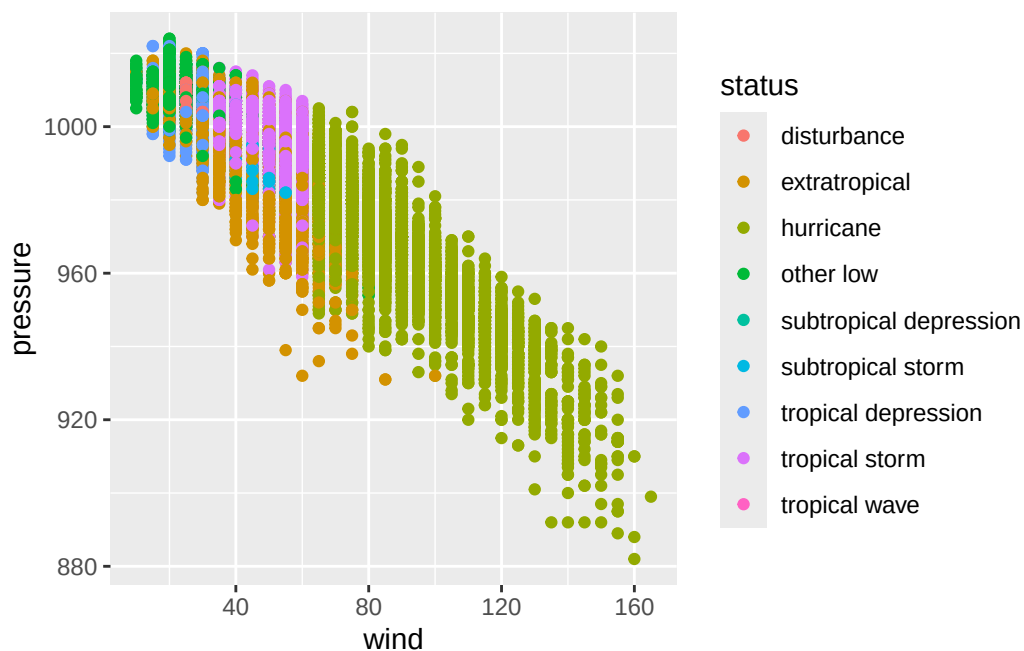
Mapping vs. setting

The single most common ggplot bug is confusing **mapping** with **setting**.

Mapping: a variable controls the aesthetic.

Goes **inside** `aes()`.

```
ggplot(storms,
  aes(x = wind, y = pressure,
      color = status)) +
  geom_point()
```

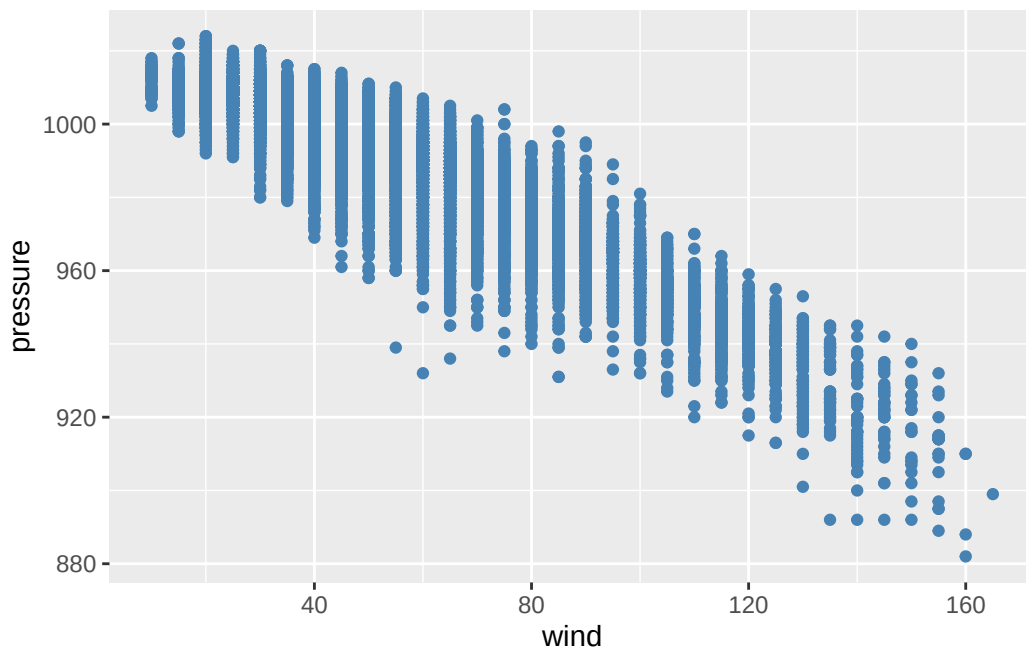


color varies by `status` value.

Setting: you fix the aesthetic to a constant.

Goes **outside** `aes()`.

```
ggplot(storms,
       aes(x = wind,
           y = pressure)) +
  geom_point(color = "steelblue")
```

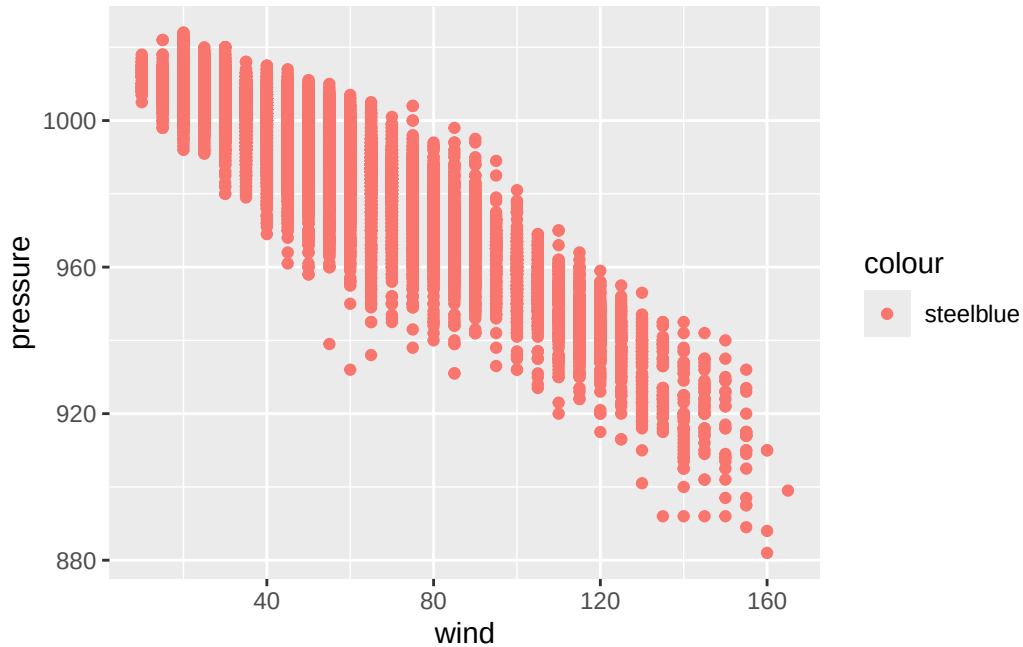


Every point is steelblue.

The classic mistake

What does this do?

```
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(aes(color = "steelblue"))
```



It makes every point a **salmon red** and adds a legend that says "**steelblue**".

Why? `aes()` says “map a variable to color.” We gave it the string "**steelblue**", which ggplot treated as a one-level categorical variable. It then picked a default color (red-pink) for that one level.

Rule of thumb: if you want a constant color, it goes *outside* `aes()`. If you want it to depend on data, it goes *inside* `aes()`.

What can you map?

Every geom accepts a subset of these aesthetics. The most common:

Table 3: Common aesthetics.

Aesthetic	Works on	Notes
x, y	almost everything	position in the frame
color	points, lines, borders	outline of a glyph
fill	bars, areas, filled shapes	interior of a glyph
size	points, lines	numerical only, ideally
alpha	almost everything	transparency, 0–1
shape	points	max 6 discrete values by default
linetype	lines	solid, dashed, dotted, ...

Aesthetic	Works on	Notes
group	grouped geoms	splits w/o adding a legend

Discrete vs. continuous aesthetics

ggplot picks a scale automatically based on the **type** of the variable.

Table 4: How ggplot scales aesthetics by variable type.

Aesthetic	Discrete variable	Continuous variable
color / fill	qualitative palette (distinct hues)	gradient (e.g., light → dark)
size	warns (order is implied but levels aren't ordered)	smooth size range
shape	distinct shapes (max 6)	error (shapes don't have an order)
alpha	warns, same reason as size	smooth

⚠ Warning

Mapping a categorical variable like `status` to `size` or `alpha` tells the reader “these categories are ordered”, which is often not actually the case. Stick to color, fill, shape, or facets for categorical splits.

Shape reference

In `ggplot`, you specify the shape of a point as a number.

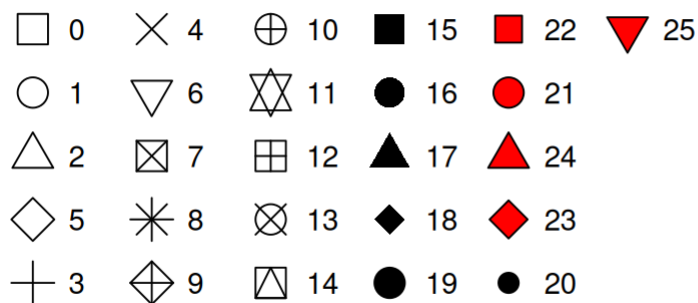
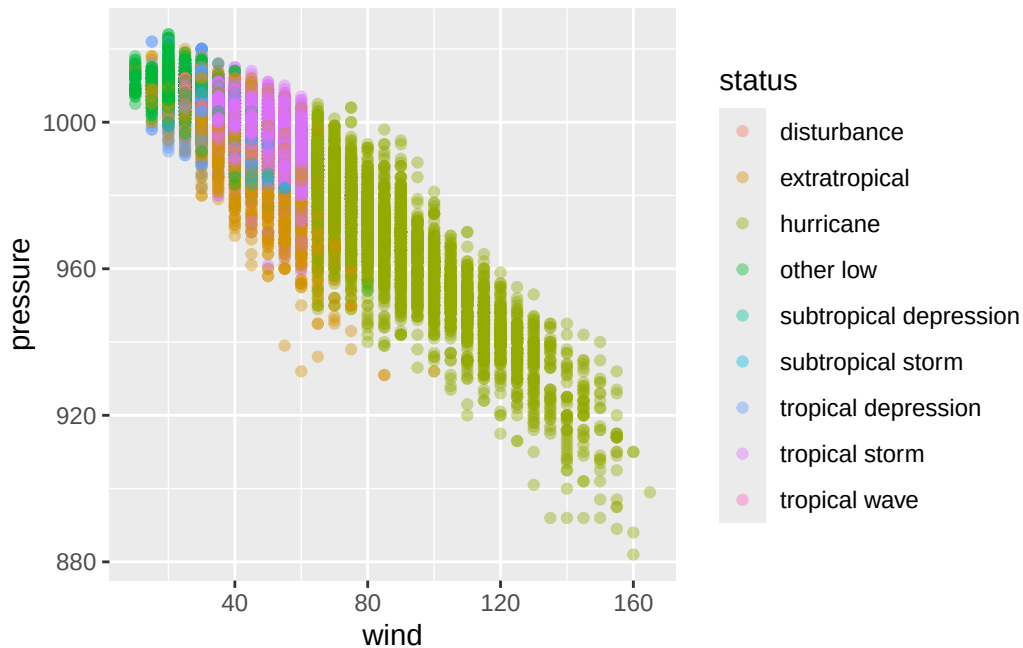


Figure 1: R has 26 built-in shapes that are identified by numbers.

Example: status mapped four ways

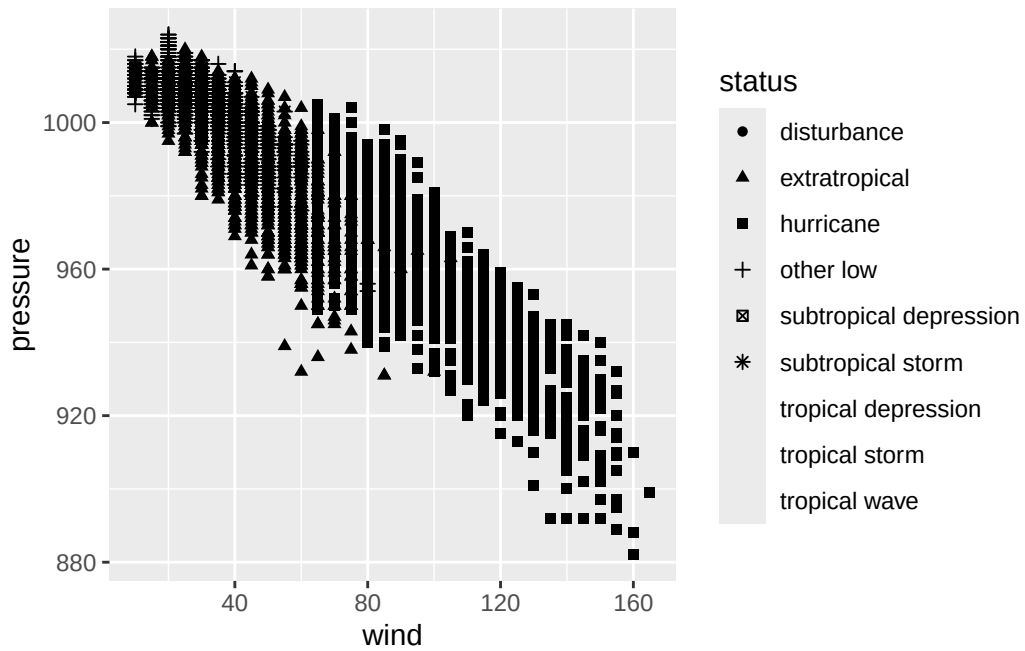
```
# Color: good for categorical
ggplot(storms, aes(x = wind, y = pressure, color = status)) + geom_point(alpha = 0.4)
```



```
# Shape: also good, but watch the 6-value limit
ggplot(storms, aes(x = wind, y = pressure, shape = status)) + geom_point()
```

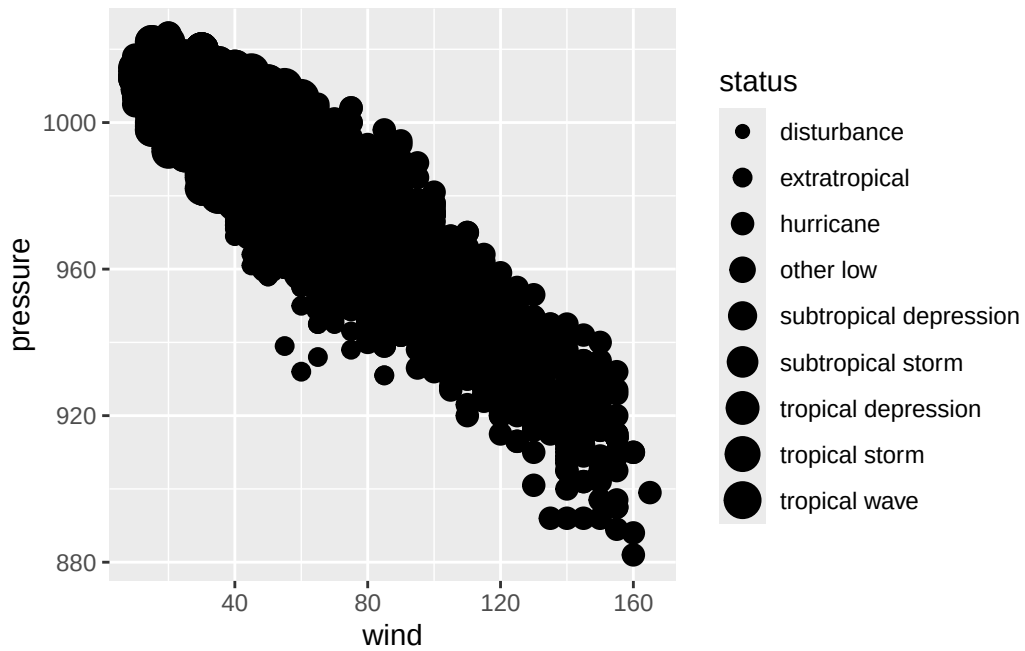
Warning: The shape palette can deal with a maximum of 6 discrete values because more than 6 becomes difficult to discriminate
i you have requested 9 values. Consider specifying shapes manually if you need that many of them.

Warning: Removed 11048 rows containing missing values or values outside the scale range (`geom\_point()`).



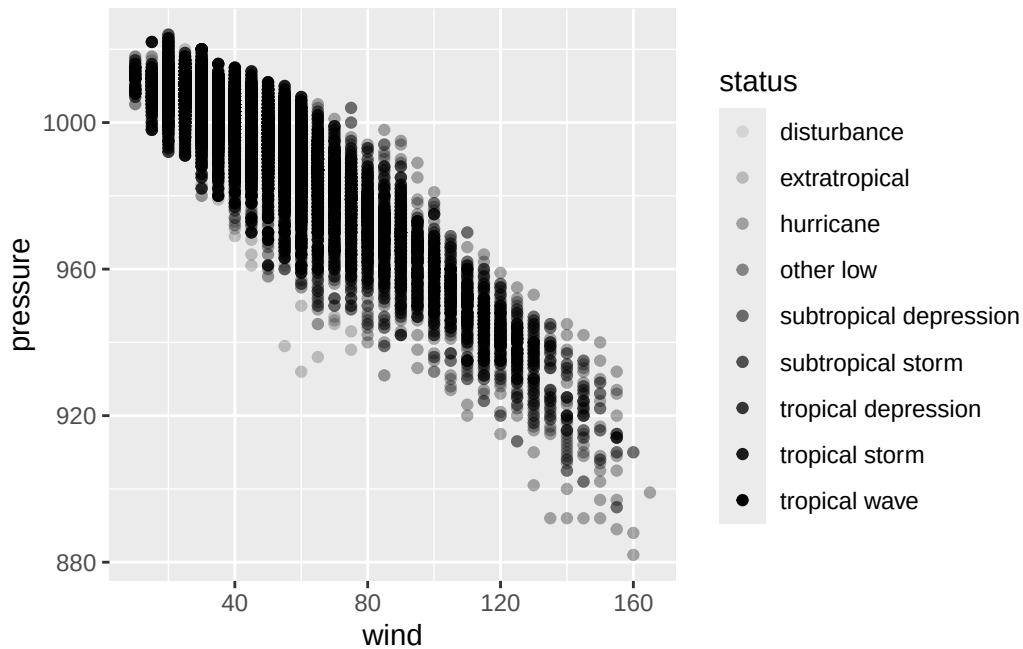
```
# Size: bad: implies ranking
ggplot(storms, aes(x = wind, y = pressure, size = status)) + geom_point()
```

Warning: Using size for a discrete variable is not advised.



```
# Alpha: same problem as size
ggplot(storms, aes(x = wind, y = pressure, alpha = status)) + geom_point()
```

Warning: Using alpha for a discrete variable is not advised.

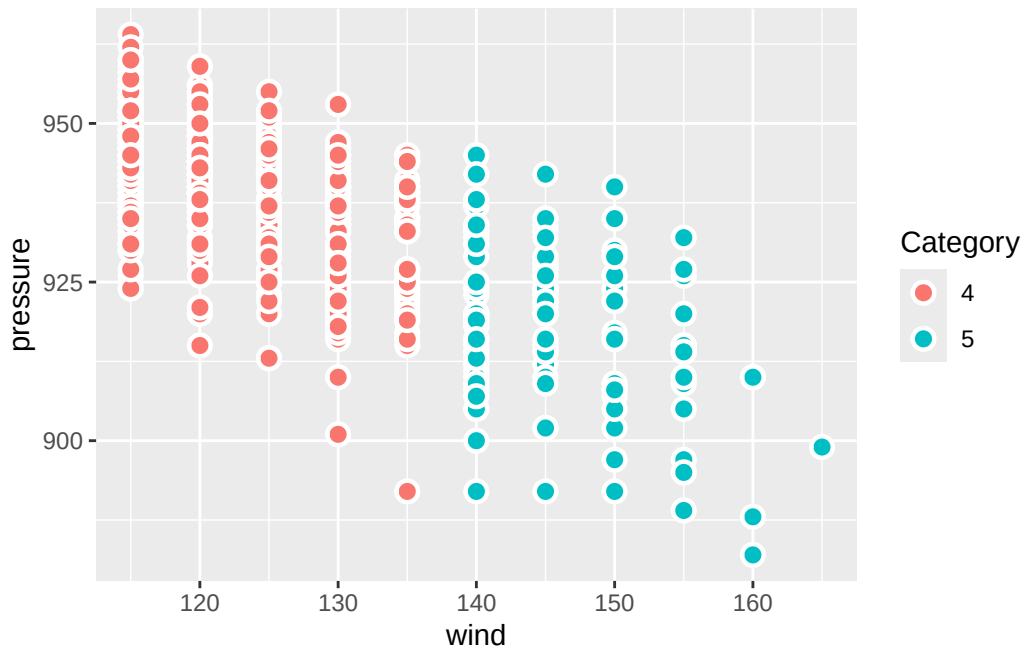


Using size/alpha for a discrete variable is not advised.

The stroke aesthetic

For filled point shapes (21–25), **stroke** controls border thickness independently of **size**.

```
ggplot(storms |> filter(category >= 4),
  aes(x = wind, y = pressure, fill = factor(category))) +
  geom_point(shape = 21, size = 3, stroke = 1.2, color = "white") +
  labs(fill = "Category")
```

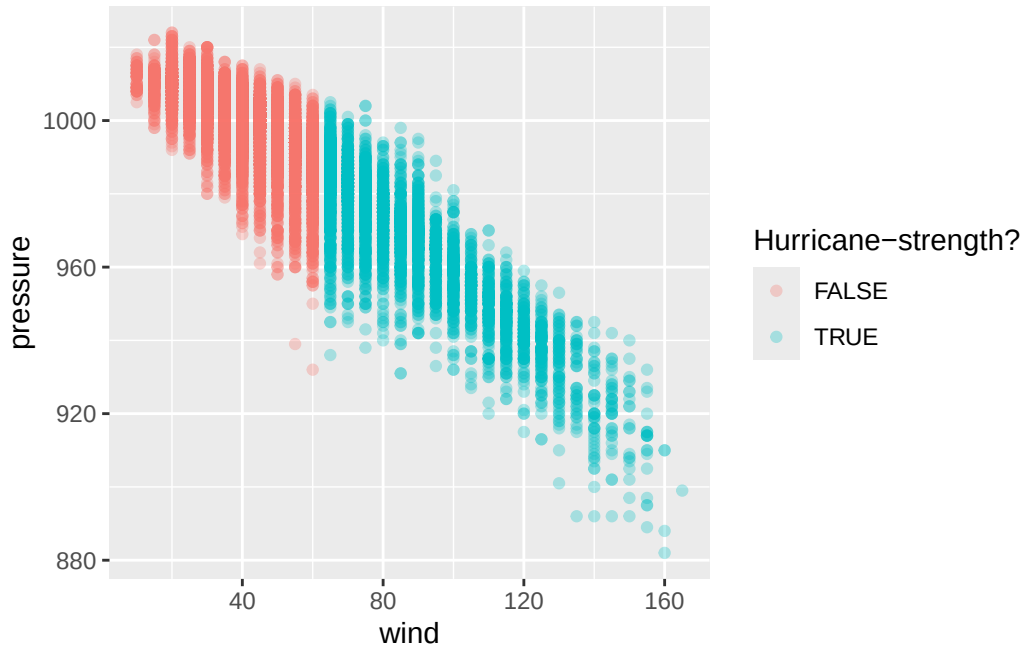


A white stroke around colored points is one of the easiest ways to make a busy plot readable.

On-the-fly transformations

Aesthetics don't have to map to a variable name; they can map to *any expression* involving variables.

```
ggplot(storms,
  aes(x = wind, y = pressure, color = wind >= 64)) +
  geom_point(alpha = 0.3) +
  labs(color = "Hurricane-strength?")
```

The expression `wind >= 64` returns `TRUE`/`FALSE` for each row, and that gets mapped to color. No need to create the column in advance.

Geoms in depth

Geoms = glyph families

Each geom produces one *type* of glyph. The big ones:

Table 5: Common geoms.

Geom	Glyph	Best for
<code>geom_point()</code>	dot	scatter
<code>geom_line()</code>	connected segments	trends over an ordered x
<code>geom_smooth()</code>	one curve (per group)	summarize a trend
<code>geom_bar()</code> / <code>geom_col()</code>	rectangles	counts / values by category
<code>geom_histogram()</code>	rectangles (bins)	distribution of one numeric var
<code>geom_boxplot()</code>	boxes + whiskers	distribution by group
<code>geom_density()</code>	smooth curve	distribution, smoothed
<code>geom_polygon()</code>	filled shapes	maps, custom regions

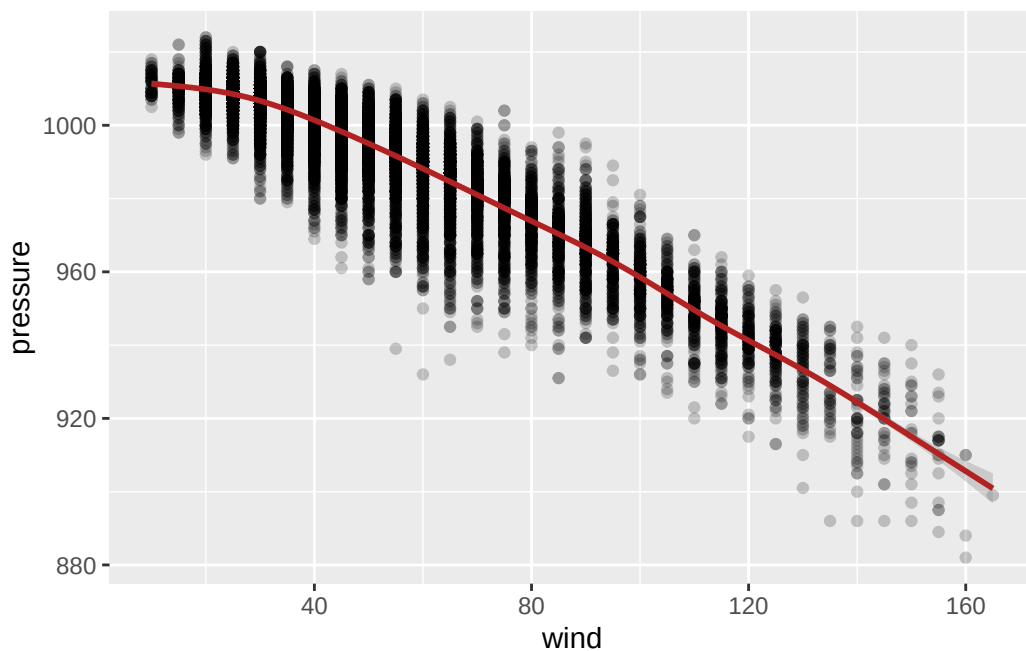
ggplot2 ships with 40+. Extension packages add hundreds more.

Layering: more than one geom

A plot can have as many geoms as you want. They stack in the order you write them.

```
ggplot(storms, aes(x = wind, y = pressure)) +  
  geom_point(alpha = 0.2) +  
  geom_smooth(color = "firebrick")
```

`geom_smooth()` using `method = 'gam'` and `formula = 'y ~ s(x, bs = "cs")'`



Here we have two layers, the same data, and the same mappings. The points show every observation; the smooth shows the trend.

i Note

The relationship between wind and pressure in tropical cyclones is *very* strong. The scatter shows what that law looks like in real storm data.

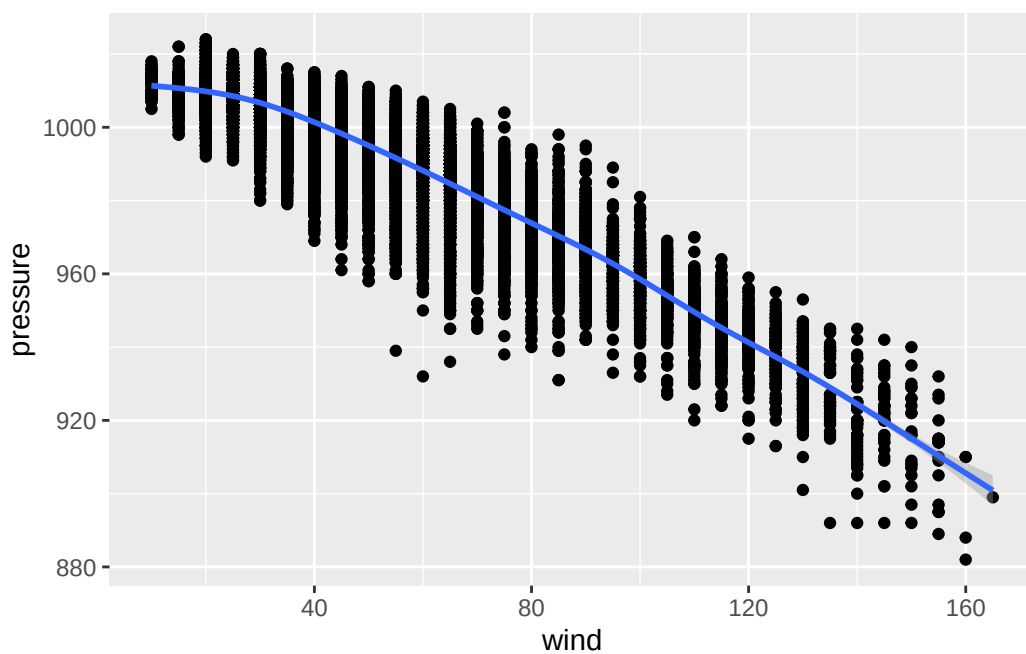
Global vs. local mappings

You can put `aes()` in two places:

In `ggplot()`: global. Every layer inherits these mappings.

```
ggplot(storms,  
       aes(x = wind, y = pressure)) +  
  geom_point() +  
  geom_smooth()
```

``geom_smooth()`` using `method = 'gam'` and `formula = 'y ~ s(x, bs = "cs")'`

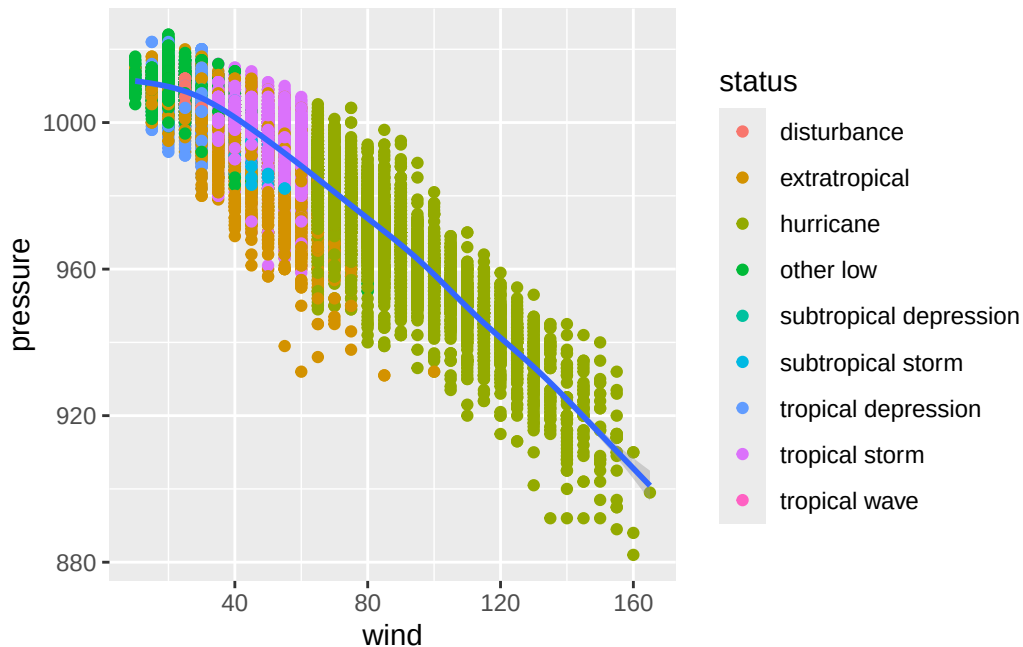


Both geoms see x and y.

In a geom: local. Only that layer sees it.

```
ggplot(storms,  
       aes(x = wind, y = pressure)) +  
  geom_point(aes(color = status)) +  
  geom_smooth()
```

``geom_smooth()`` using `method = 'gam'` and `formula = 'y ~ s(x, bs = "cs")'`



Only points are colored; the smooth is one curve.

Why local mappings matter

If you put `color = status` in the global `aes()`, **every layer** sees it. `geom_smooth()` will draw one smoother *per status*, not one overall.

```
# One smoother per status (probably not what you want here):
ggplot(storms, aes(x = wind, y = pressure, color = status)) +
  geom_point(alpha = 0.2) +
  geom_smooth()
```

``geom_smooth()`` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: Failed to fit group 1.

Caused by error in ``smooth.construct.cr.smooth.spec()``:

! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 5.

Caused by error in ``smooth.construct.cr.smooth.spec()``:

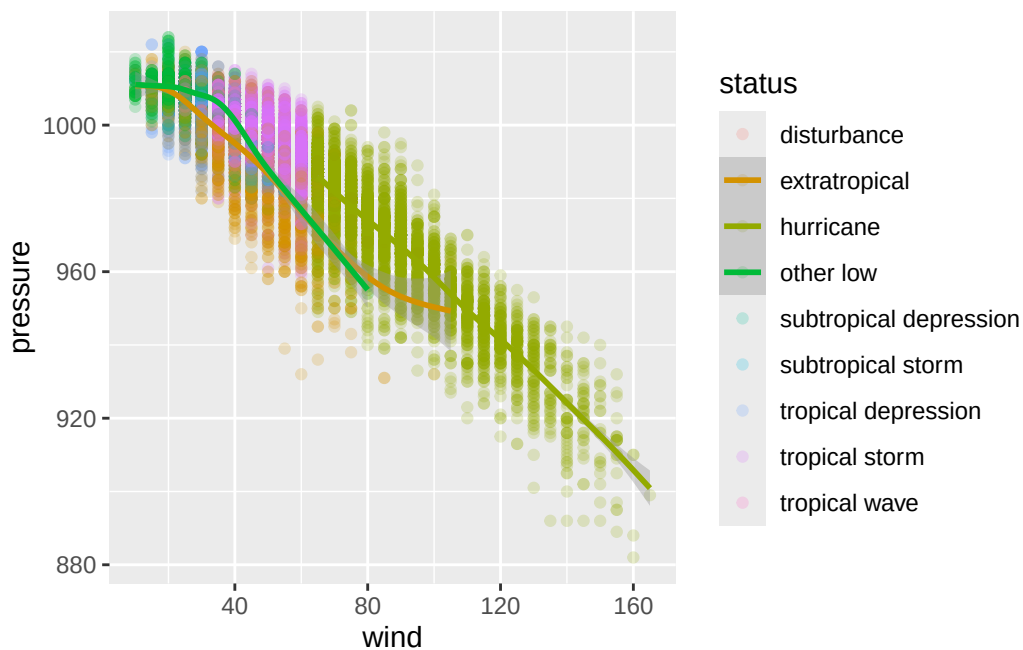
! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 6.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 7.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.

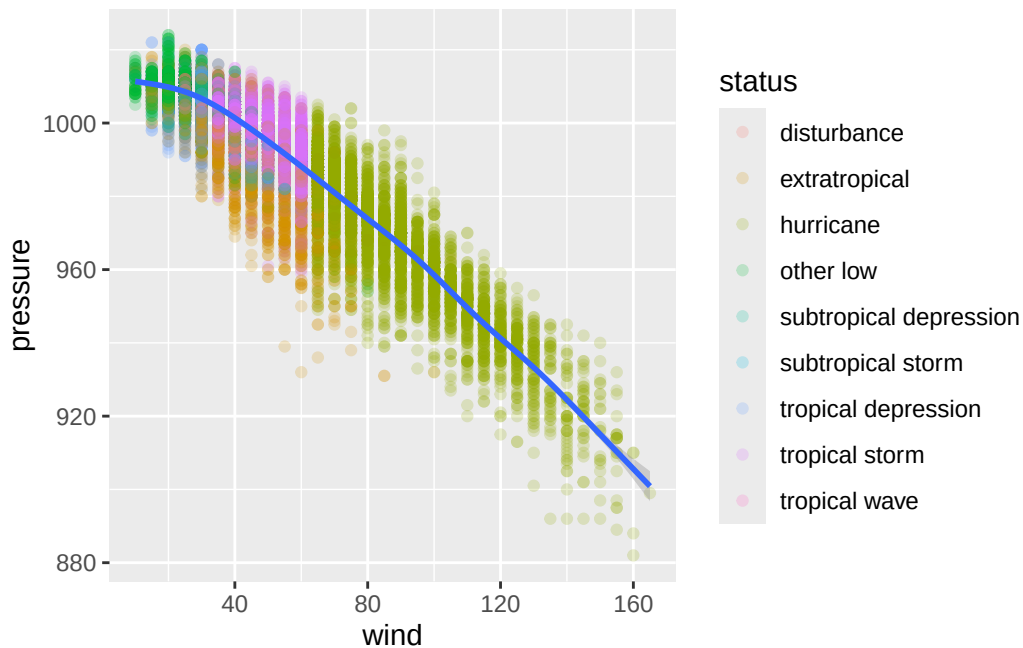
Warning: Failed to fit group 8.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 9.
Caused by error in `smooth.construct.cr.smooth.spec()`:
! x has insufficient unique values to support 10 knots: reduce k.



```
# Points colored by status, one global smoother:  
ggplot(storms, aes(x = wind, y = pressure)) +  
  geom_point(aes(color = status), alpha = 0.2) +  
  geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Where you put the mapping changes the plot. There's no "right" choice (just be *intentional* in your choices.)

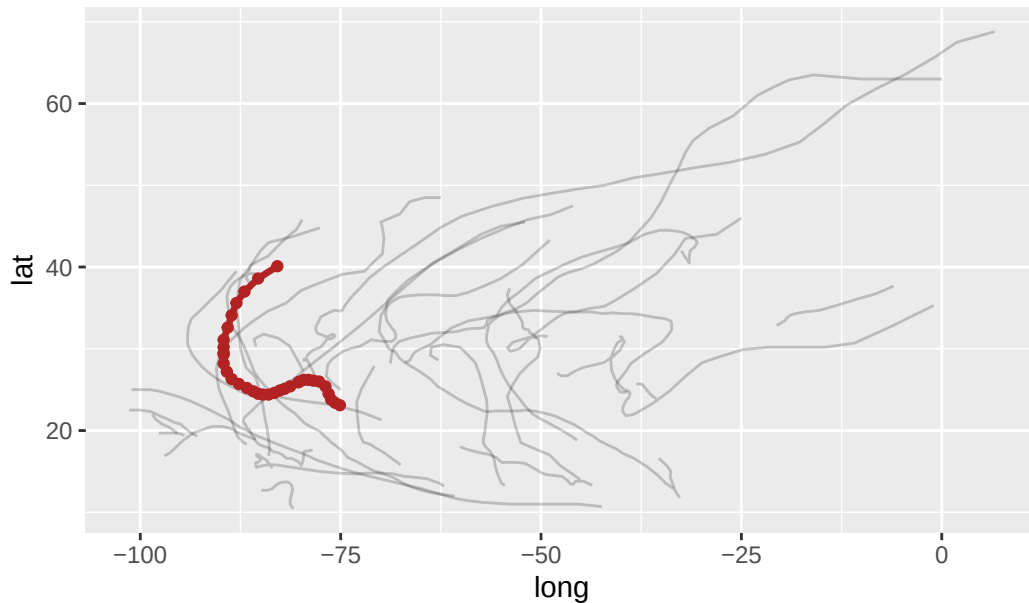
Different data per layer

Each layer can have its **own data**. This is how you highlight a subset.

```
katrina <- storms |> filter(name == "Katrina", year == 2005)

ggplot(storms |> filter(year == 2005),
       aes(x = long, y = lat)) +
  geom_path(aes(group = name), alpha = 0.2) +
  geom_path(data = katrina, color = "firebrick", linewidth = 1.2) +
  geom_point(data = katrina, color = "firebrick") +
  labs(title = "2005 Atlantic hurricane season, with Katrina highlighted")
```

2005 Atlantic hurricane season, with Katrina highlighted



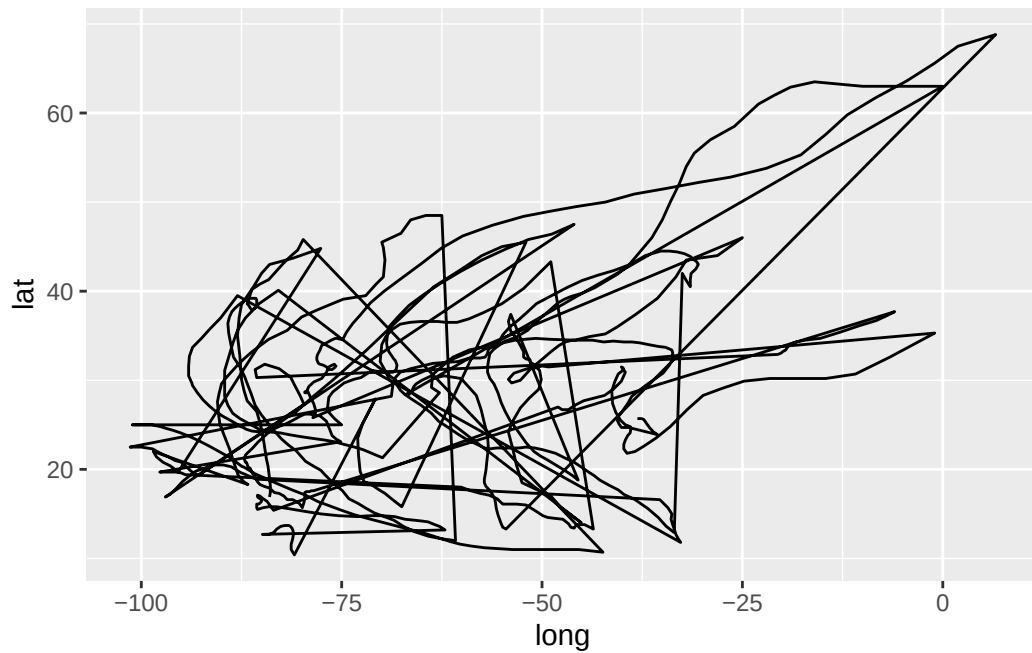
Three layers: all storms (faint), Katrina's path (red), Katrina's observations (red dots). Each calls `data =` locally to override the default.

The group aesthetic

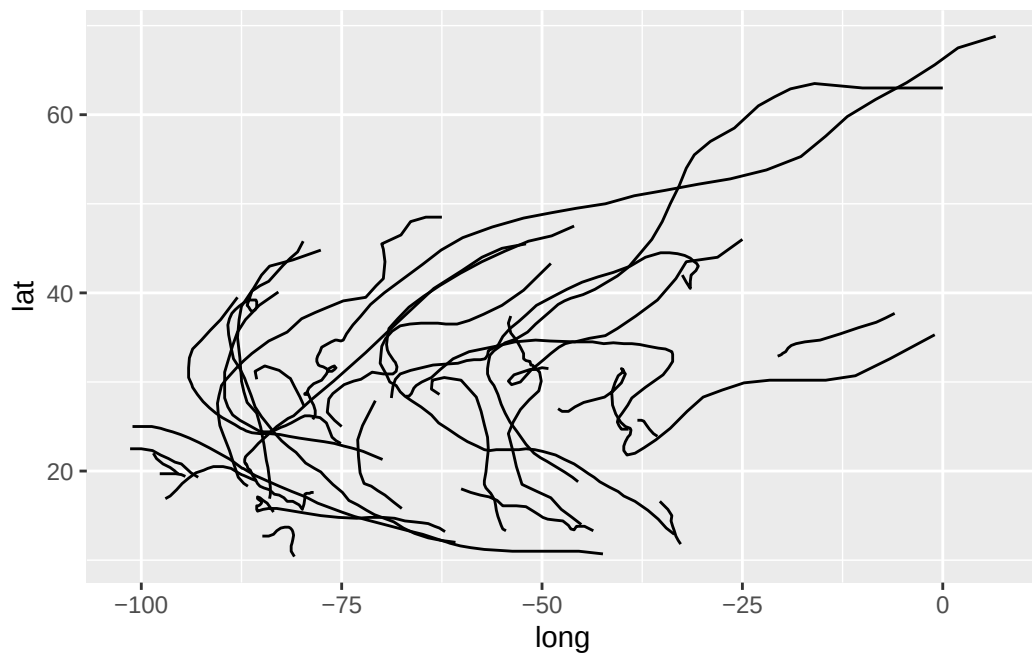
Some geoms — lines, smoothers, polygons — connect multiple rows. They need to know **which rows belong together**.

If you map a discrete variable to color or linetype, ggplot groups for you automatically. If you don't, you may need to set `group` explicitly.

```
# Bad: one giant line zigzagging between every storm
ggplot(storms |> filter(year == 2005), aes(x = long, y = lat)) +
  geom_path()
```



```
# Good: one path per storm
ggplot(storms |> filter(year == 2005), aes(x = long, y = lat, group = name)) +
  geom_path()
```



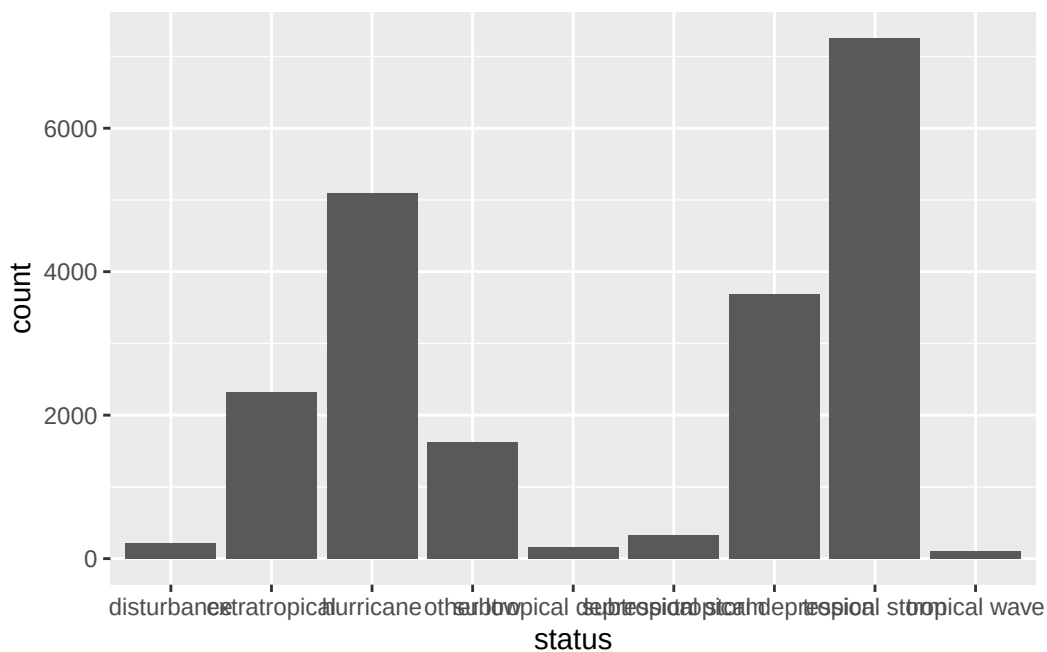
`group` doesn't add a legend or change colors, it just splits the geom.

Statistical transformations

Where do bar heights come from?

Here's a (potentially) tricky question:

```
ggplot(storms, aes(x = status)) +  
  geom_bar()
```



We told `ggplot` about `x = status`. Where did the y-axis (`count`) come from?

It was computed. `geom_bar()` doesn't just plot raw data. It also *transforms* it by grouping rows by `status` and counting them, then plotting the counts.

Every geom has a **stat** (statistical transformation) it runs under the hood.

Every geom ↔ stat pair

Table 6: Geoms and their default stats.

Geom	Default stat	What it computes
<code>geom_bar()</code>	<code>stat_count()</code>	counts per x value
<code>geom_histogram()</code>	<code>stat_bin()</code>	counts per bin
<code>geom_boxplot()</code>	<code>stat_boxplot()</code>	five-number summary
<code>geom_smooth()</code>	<code>stat_smooth()</code>	fitted curve + CI
<code>geom_density()</code>	<code>stat_density()</code>	kernel density estimate
<code>geom_point()</code>	<code>stat_identity()</code>	nothing — just plots raw

You can use a geom and never think about its stat. But sometimes you will need to think about it.

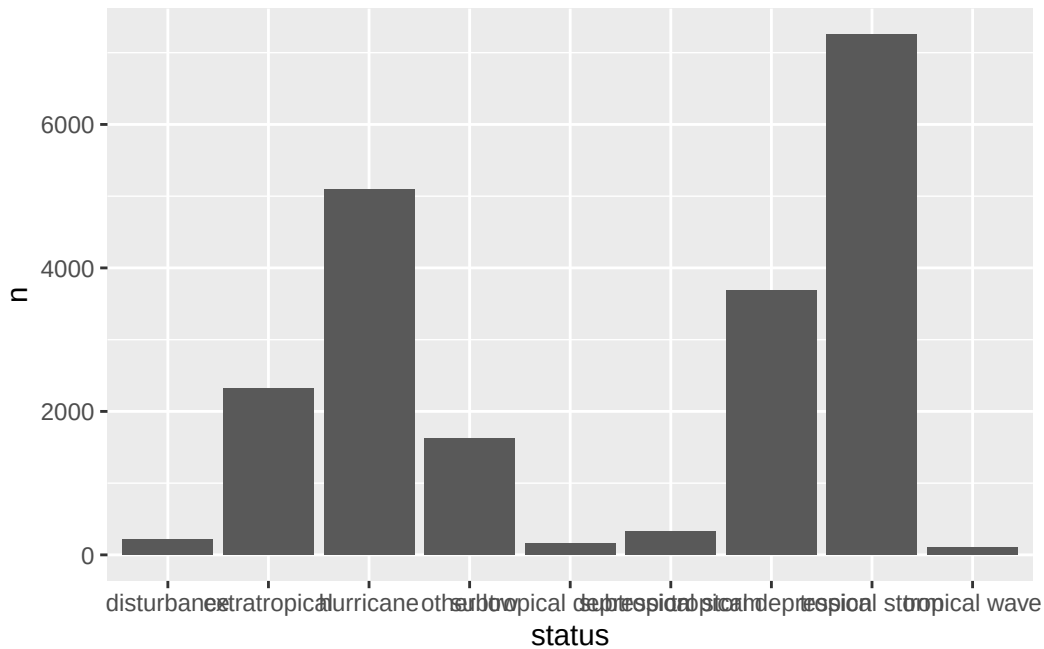
Three reasons to touch the stat

1. **You already pre-computed the values.** We can override the default behavior of `geom_bar` with `stat = "identity"`.

```
storm_counts <- storms |> count(status)
storm_counts
```

```
# A tibble: 9 x 2
  status          n
  <fct>        <int>
1 disturbance    212
2 extratropical 2318
3 hurricane     5100
4 other low     1623
5 subtropical depression 154
6 subtropical storm   323
7 tropical depression 3685
8 tropical storm   7252
9 tropical wave    111
```

```
ggplot(storm_counts, aes(x = status, y = n)) +
  geom_bar(stat = "identity") # or just use geom_col(), it's the same thing
```



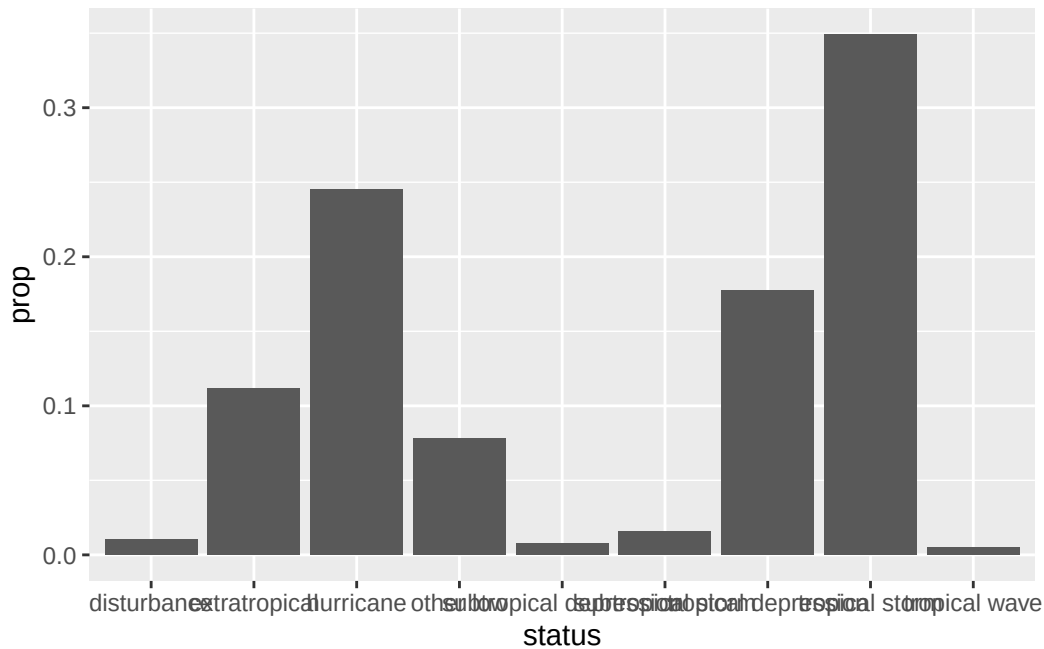
`geom_col()` is a shortcut for `geom_bar(stat = "identity")`. You can use it when you have your own y values.

Three reasons to touch the stat

2. You want a *computed* variable, not the default.

`stat_count()` actually computes two things: `count` (default) and `prop` (proportion). You can ask for the proportion using `after_stat()`:

```
ggplot(storms, aes(x = status, y = after_stat(prop), group = 1)) +
  geom_bar()
```



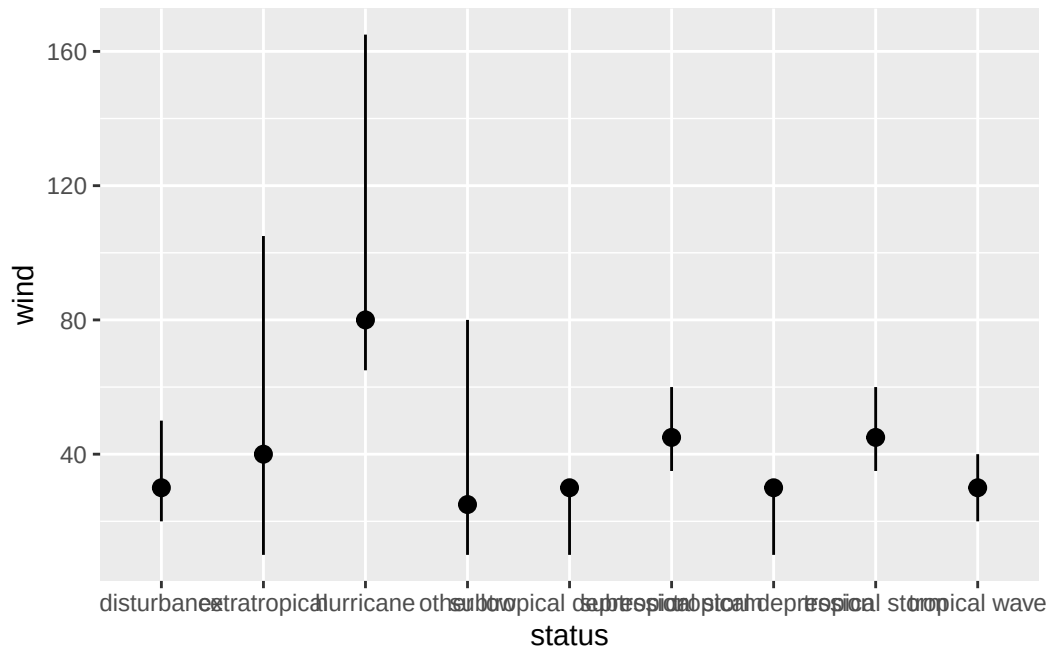
⚠ Why group = 1?

By default, `prop` is computed *within each x group*, which means every bar would be 100%. Setting `group = 1` tells ggplot “treat all rows as one group when computing proportions” so they sum to 1 across the whole plot.

Three reasons to touch the stat

3. You want to make the transformation explicit. Use a `stat_*`() function directly.

```
ggplot(storms, aes(x = status, y = wind)) +
  stat_summary(
    fun.min = min,
    fun.max = max,
    fun = median
  )
```

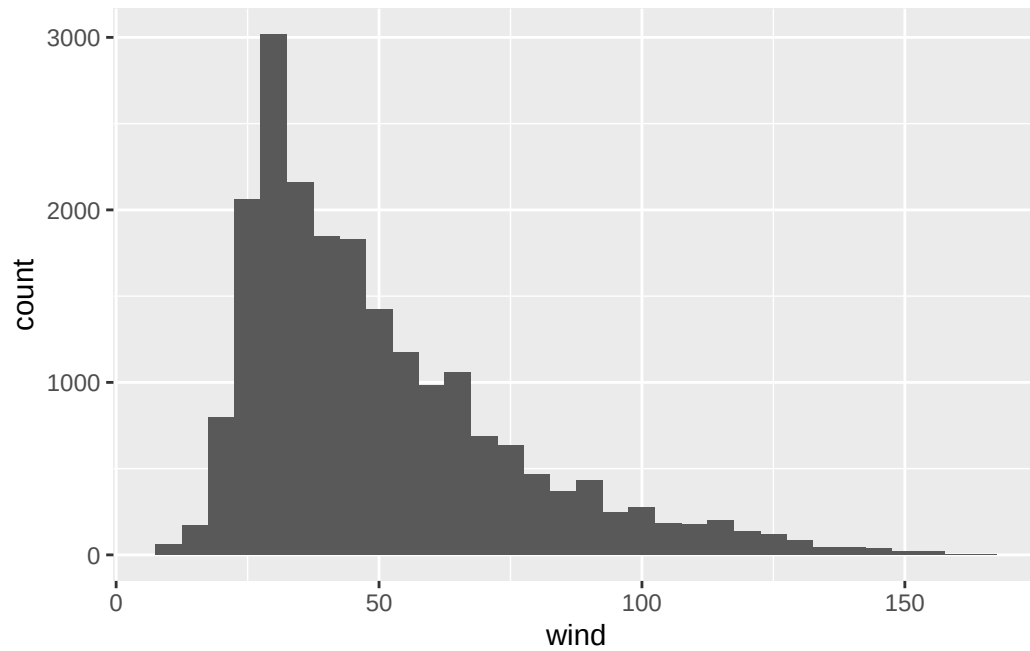


This shows the min, median, and max wind speed for each status without you having to manually write any `dplyr` code first. `stat_summary()` does the aggregation as part of the plot.

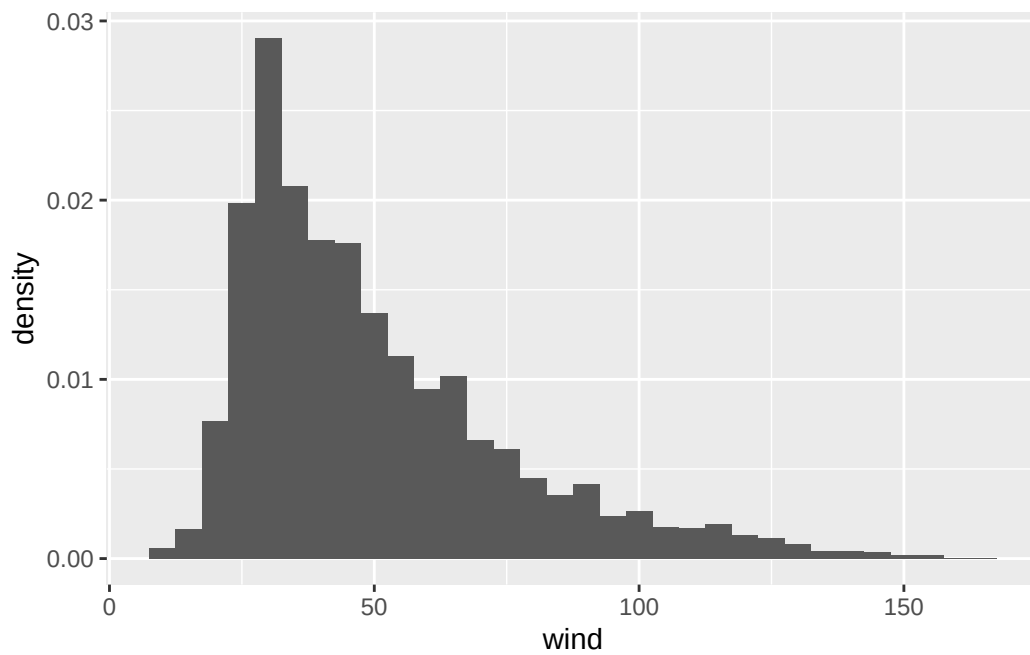
`after_stat()` quick overview

Anywhere you'd write a variable name inside `aes()`, you can wrap it in `after_stat(...)` to refer to a **computed** variable instead of one in the original data.

```
# Default: bin counts on y
ggplot(storms, aes(x = wind)) +
  geom_histogram(binwidth = 5)
```



```
# Same bins, but density on y instead  
ggplot(storms, aes(x = wind, y = after_stat(density))) +  
  geom_histogram(binwidth = 5)
```



To see what's available for a given geom, run `?geom_bar` and look for the **Computed variables** section.

Position adjustments

What's a position adjustment?

When multiple glyphs would land in the same spot, ggplot has to decide what to do. That decision is the **position adjustment**.

The big four:

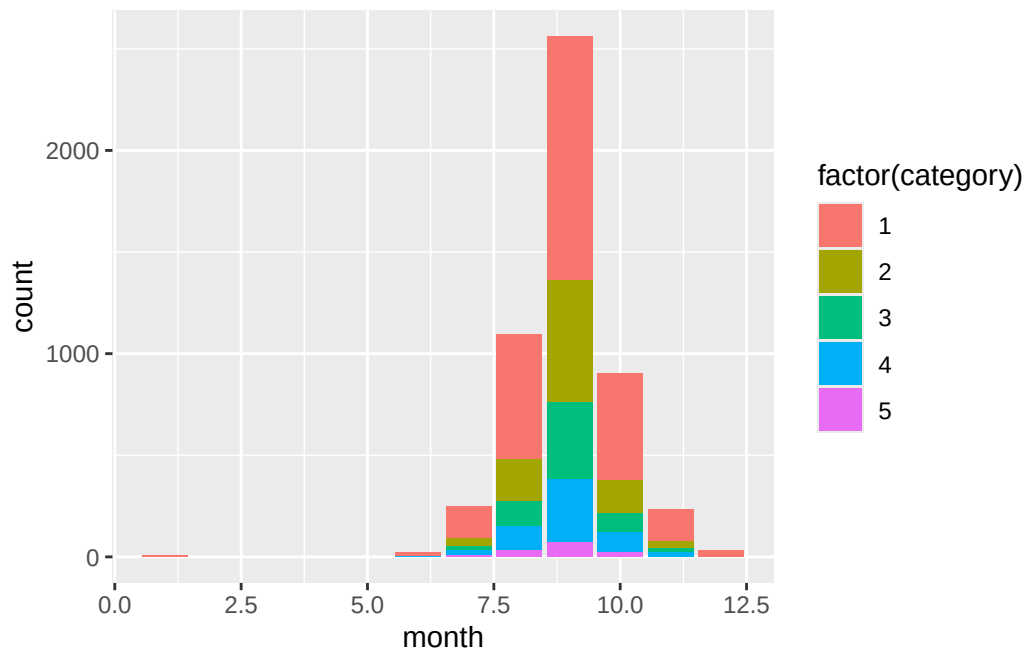
Table 7: ggplot's main position adjustments.

Position	What it does	Default for
"identity"	leave it alone	points, lines
"stack"	stack on top of each other	bars (when filled by a 2nd var)
"dodge"	side-by-side	(you have to ask)
"jitter"	add small random noise	(you have to ask)
"fill"	stack and normalize to 1	(you have to ask)

Stacking (the default for filled bars)

```
hurricanes <- storms |> filter(status == "hurricane")

ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar()
```



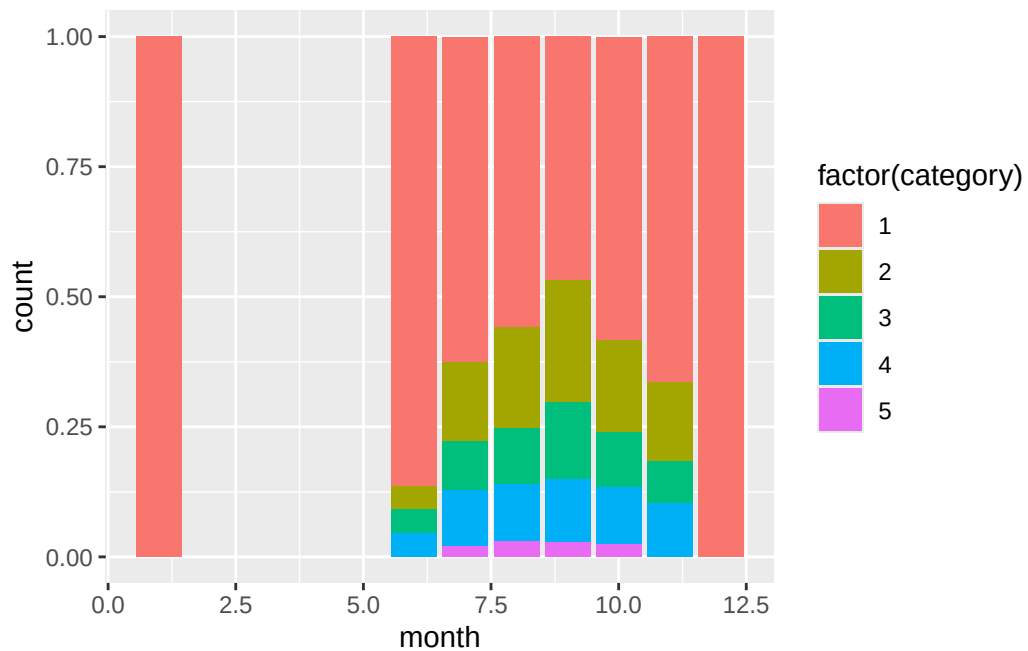
Each month gets one bar; segments stack by hurricane category.

This is useful for showing total *and* composition at once, but hard to compare middle segments across bars (their bottoms aren't aligned).

position = "fill"

Here we have the same code, but every bar is normalized to height 1:

```
ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar(position = "fill")
```

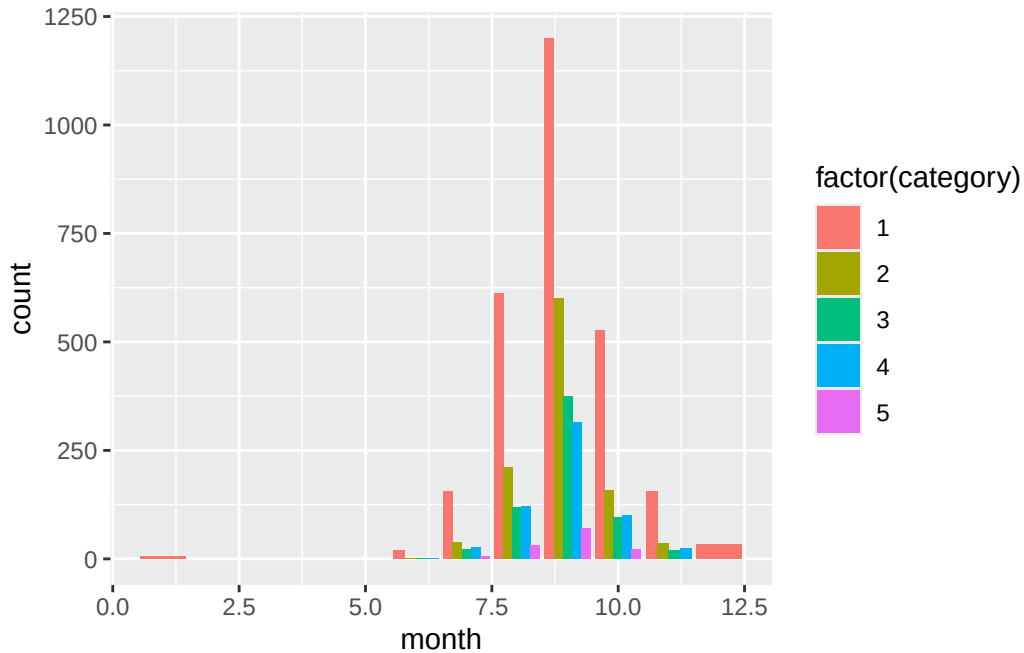



Now the y-axis is proportion, not count. Loses information about total volume, but makes the *composition* easy to compare across months.

position = "dodge"

Here are the bars side-by-side instead of stacked:

```
ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar(position = "dodge")
```



This is the easiest version for comparing individual counts directly. The trade-off: lots of bars, can get crowded fast.

💡 Quick rule

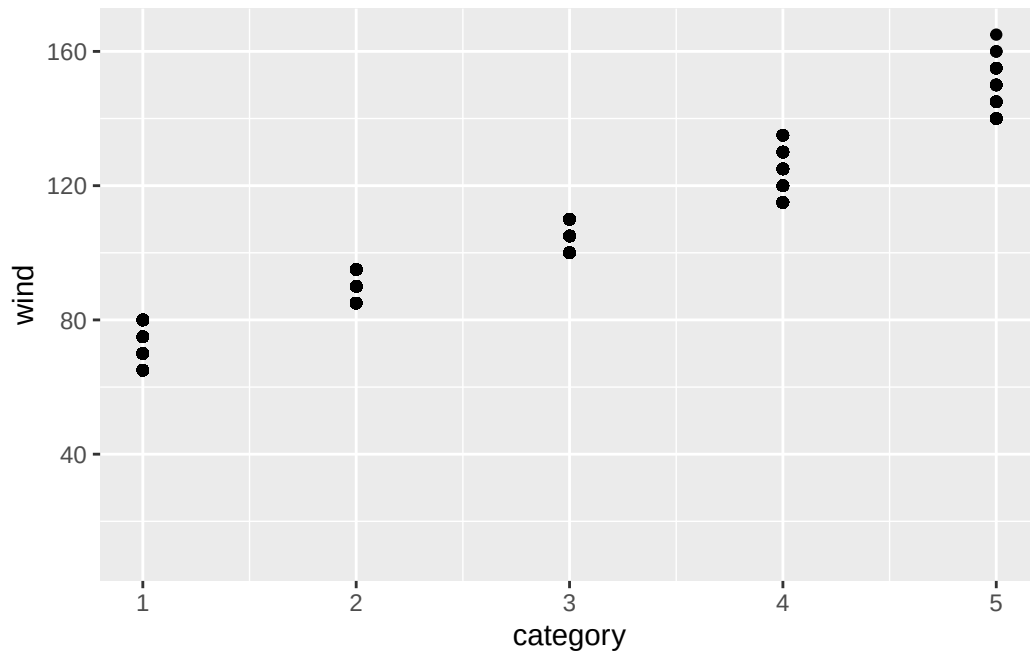
- **Stack:** show totals, with composition as bonus.
- **Fill:** show composition only, ignore totals.
- **Dodge:** compare individual values, ignore totals.

Jittering — for scatterplots, not bars

A different problem: many storm observations are recorded only every 6 hours, and pressure readings are rounded to whole millibars. Lots of points overlap.

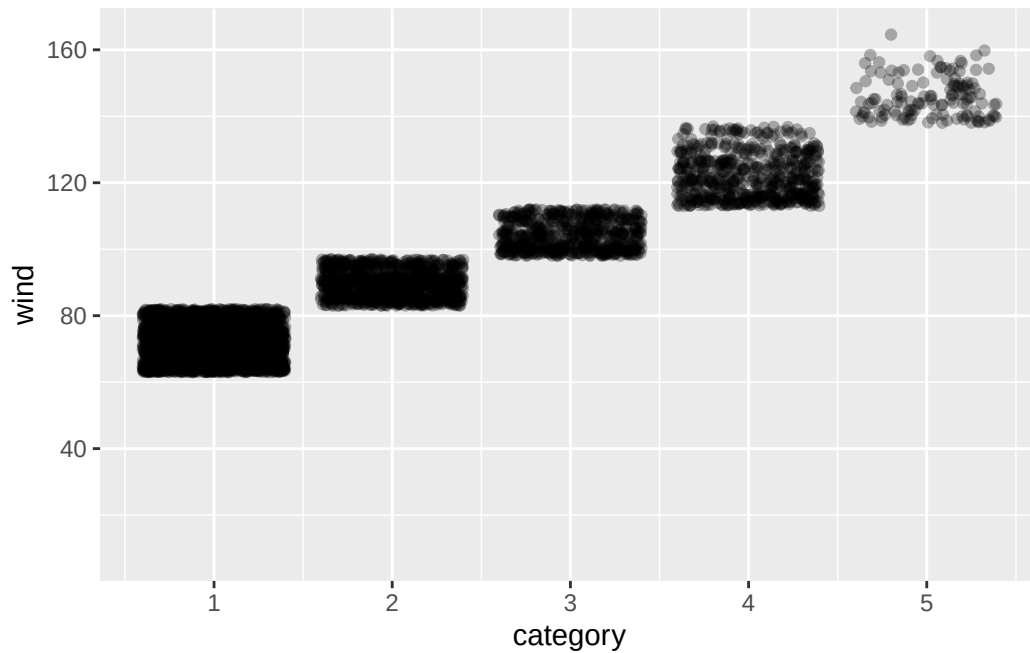
```
# Lots of overplotting since points are stacked on top of each other
ggplot(storms, aes(x = category, y = wind)) +
  geom_point()
```

Warning: Removed 15678 rows containing missing values or values outside the scale range (`geom\_point()`).



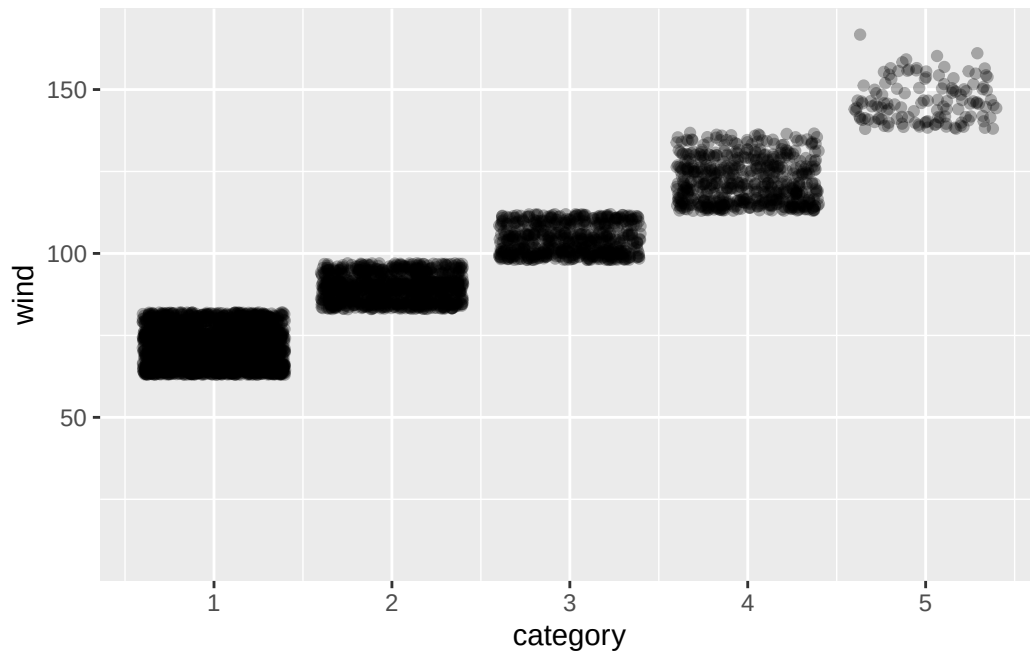
```
# Spread them out a bit
ggplot(storms, aes(x = category, y = wind)) +
  geom_point(position = "jitter", alpha = 0.3)
```

Warning: Removed 15678 rows containing missing values or values outside the scale range (`geom\_point()`).



```
# Shorthand for the same thing
ggplot(storms, aes(x = category, y = wind)) +
  geom_jitter(alpha = 0.3)
```

Warning: Removed 15678 rows containing missing values or values outside the scale range (``geom_point()``).



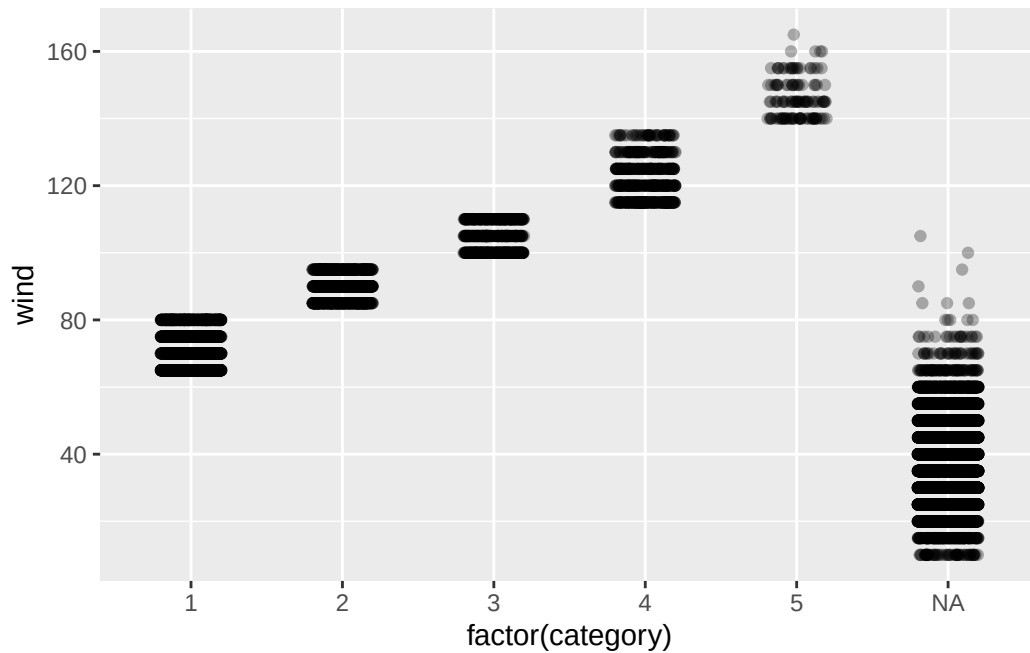
Jitter adds tiny random noise so points stop landing on top of each other. You lose precision but (often) can gain a clear picture of density.

Don't over-jitter

`geom_jitter()` takes `width` and `height` arguments. Defaults are usually fine, but:

- **Categorical x:** small `width` (~0.2), `height` = 0 to keep y meaningful.
- **Integer x and y:** small jitter on both.
- **Continuous data:** you probably don't need jitter at all, just use `alpha` or `geom_hex()`/`geom_bin2d()` instead.

```
ggplot(storms, aes(x = factor(category), y = wind)) +  
  geom_jitter(width = 0.2, height = 0, alpha = 0.3)
```



The big picture

The layered grammar of graphics

Putting it all together, *every* ggplot has the same skeleton:

```
ggplot(data = <DATA>) +  
  <GEOM_FUNCTION>(  
    mapping = aes(<MAPPINGS>),  
    stat = <STAT>,  
    position = <POSITION>  
  ) +  
  <COORDINATE_FUNCTION> +  
  <FACET_FUNCTION>
```

There are seven slots here. You only want to fill the ones you need to change since ggplot picks sensible defaults for the rest.

That's why just `ggplot()` + `aes()` + `geom_*()` go a long way on their own. The other slots are there when you need them.

What we covered today

- The **vocabulary**: frame, glyph, aesthetic, scale, guide.
- **Mapping** (inside `aes()`) vs. **setting** (outside `aes()`) and the bug where you mix them up.
- Aesthetics: x, y, color, fill, size, shape, alpha, linetype, group, stroke. Discrete vs. continuous behavior.
- **Geoms**: stacking layers, global vs. local mappings, different data per layer, the **group** aesthetic.
- **Stats**: the hidden transformation in every geom; `geom_col()` vs `geom_bar()`; `after_stat()` for proportions and densities; `stat_summary()` for ad hoc aggregation.
- **Position adjustments**: identity, stack, fill, dodge, jitter — when to use which.

Activity

Activity: ggplot internals, with storms

Two parts.

- [Activity Canvas Link](#)

Part 1: Guided warmup. Five short fill-in-the-blank chunks. Practice each concept from today on a fresh question.

Part 2: Open exploration. Pick one of three guiding questions about the `storms` data and build the plot you think answers it best. We'll show a few at the end.

To turn in: Save your `.qmd` and rendered HTML.

Activity debrief

Things to watch for:

- `aes(color = "red")` instead of `color = "red"` (and getting confused when nothing turns red.)
- A `geom_path()` that zigzags wildly because `group` is missing.
- `geom_bar(stat = "identity")` when you already have counts (use `geom_col()`).
- `position = "fill"` when you actually wanted `"dodge"` (proportions vs. side-by-side counts.)
- Mapping a categorical to `size`.

Wrap-up & looking ahead

Tomorrow: facets in depth, scales, and color choices, including continuous color scales, sequential vs. diverging palettes, and accessibility.

Friday: Quiz 2 + a graphics critique activity.

Upcoming Deadlines

- **Thurs 5/28, before class:** [DC §8.1–8.3](#)
- **Fri 5/29, in class:** Quiz 2
- **Fri 5/29, 11:59 p.m.:** [HW 2: Pipes, Plots & Reading Graphics](#)

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 9](#) · [DC Ch. 6](#) · [ggplot2 reference](#)